Scan2BIM Inshelter — MVP Infrastructure Requirements

This page defines the **minimum viable infrastructure** required to run the Scan2BIM Inshelter pipeline end to end for early delivery, validation, and limited operational use. The focus is only on MVP infrastructure needed to process approximately **5–10 scans per day** at the lowest practical cost while preserving a clear upgrade path to production.

> **MVP objective:** Run the full workflow from webhook trigger to IFC delivery with one low-cost Azure-based environment, one on-demand GPU path, durable workflow tracking, secure service-to-service access, and basic operational recovery.

MVP scope

- **Included:** compute, GPU execution, orchestration, storage, messaging, database, workflow engine, image registry, ingress, secrets, and baseline security controls
- **Target throughput:** 5–10 scans per day
- **Expected concurrency:** 1–2 scans at a time
- **Expected processing time:** roughly 45–120 minutes per scan depending on scan size and GPU startup time
- **Deployment model:** single-region, single-environment MVP

> **Explicit MVP trade-off:** this design favors cost and speed of setup over redundancy, guaranteed GPU availability, and high availability. It is suitable for pilot delivery, not for business-critical production commitments.

MVP infrastructure requirements

Component	MVP requirement	Purpose	MVP note
AKS cluster	1 standard node pool with `Standard_D4s_v3` class capacity	Runs Root MS, Post-Processing, ingress, KEDA, Flux CD, and minimal Temporal services	Single-node design keeps cost low but has no node-level redundancy
GPU node pool	1 spot GPU node pool using `Standard_NC6s_v`	Runs Segmentation workload	Spot capacity may be interrupted; retry handling is required

	`3` or equivalent V100-backed capacity, scaled to zero when idle		
Container orchestration	Azure Kubernetes Service in one namespace such as `esdt-s2b`	Hosts all containerized workloads	Provides the operational base for future scale-out
Message bus	Azure Service Bus Standard with three queues	Decouples services and buffers long-running steps	Enough for MVP because messages are small
Object storage	Azure Blob Storage Hot tier, LRS, minimum 500 GB	Stores intermediate artifacts, segmented outputs, and backup IFC files	No geo-redundancy in MVP
Application database	MongoDB Atlas M10 shared tier	Stores workflow state, scan tracking, deduplication, and operational metadata	Adequate for early low-volume usage
Workflow persistence	Temporal self-hosted with PostgreSQL backend on Azure Database for PostgreSQL Flexible Server Burstable B2s	Provides durable orchestration for long-running jobs	Minimal deployment footprint for MVP
Image registry	Azure Container Registry Basic	Stores deployable container images	Sufficient for one environment and limited pull volume
Ingress	NGINX Ingress Controller with TLS	Accepts webhook traffic into Root MS	Single external entry point for MVP
Certificate management	cert-manager with DNS-managed certificates	Automates HTTPS certificate issuance and renewal	Required for secure external webhook access

Secrets management	Azure Key Vault plus SOPS-encrypted configuration	Protects credentials and integration secrets	Supports secure GitOps workflow
Deployment automation	Flux CD	Applies Kubernetes manifests from Git	Improves repeatability and upgrade path
Event-driven scaling	KEDA for queue-based scaling of segmentation jobs	Turns GPU compute on only when work exists	Critical for MVP cost control

Minimum service topology

- **Root Microservice:** 1 replica on the standard AKS node
- **Post-Processing Microservice:** 1 replica on the standard AKS node
- **Segmentation Microservice:** 0–1 replica on the GPU pool, started on demand
- **Temporal:** minimal self-hosted deployment sized only for MVP workflow volume
- **NGINX Ingress, KEDA, Flux CD, cert-manager:** shared cluster services on the standard node

> **Recommended MVP principle:** keep all always-on services on one standard node and isolate GPU cost to the segmentation step only. This is the single biggest cost control lever in the design.

Required queues and storage structure

The MVP needs a simple but durable asynchronous backbone.

- **Service Bus queues**
  - `esdt-s2b-root-queue`
  - `esdt-s2b-segmentation-queue`
  - `esdt-s2b-post-processing-queue`
- **Dead-letter support** must be enabled through native queue behavior for failed messages
- **Blob container** such as `segmentation-store` for segmented outputs, temporary artifacts, and IFC backup copies

This is enough to support decoupled processing, retry behavior, and manual recovery of failed jobs.

Security requirements for MVP

Security area	MVP requirement	Reason
Transport security	HTTPS on external ingress and TLS for external integrations	Protects webhook and API traffic
Azure resource access	Managed Identity / Workload Identity where supported	Avoids static credentials in code or config
Secrets	Store integration secrets in Key Vault and encrypted Git-managed files via SOPS	Supports safe deployment and rotation
Kubernetes access control	Basic RBAC with least privilege service accounts	Limits blast radius inside the cluster
External DataRepo access	AWS Cognito token flow for machine-to-machine authentication	Required integration dependency outside Azure
Network posture	No public IPs on application pods; ingress only through NGINX	Reduces exposed attack surface

MVP non-functional requirements

- **Availability target:** approximately 95% is acceptable for MVP
- **Recovery model:** service restart and queue retry first; manual intervention acceptable for exceptional failures
- **Data durability:** durable workflow and state tracking are required even if compute restarts
- **Scalability:** architecture must scale to production by increasing node counts and service tiers rather than redesigning components

- **Observability:** application logs and workflow state must be retained in a way that allows debugging of failed scans

Known MVP limitations

- SINGLE REGION No disaster recovery across regions
- SINGLE STANDARD NODE No high availability for always-on services
- SPOT GPU Segmentation jobs may be interrupted and restarted
- SHARED DATABASE TIER Limited headroom for larger operational volume
- STANDARD SERVICE BUS Lower enterprise networking and isolation features than production

Acceptance criteria for MVP infrastructure

- [ ] One AKS environment is deployed with a standard node pool and a scale-to-zero GPU node pool
- [ ] Root MS, Segmentation MS, and Post-Processing MS can be deployed from ACR through Flux CD
- [ ] Temporal is running with PostgreSQL-backed durable state
- [ ] Service Bus queues support end-to-end job handoff and retry
- [ ] Blob Storage supports intermediate outputs and IFC backup storage
- [ ] External webhook can reach the Root MS through NGINX over HTTPS
- [ ] Segmentation can trigger GPU scale-up from queue activity and scale back to zero when idle
- [ ] Post-Processing can authenticate to DataRepo through Cognito and upload final IFC output
- [ ] Failed messages are recoverable through retry and dead-letter handling
- [ ] No hard-coded passwords are required in application code or deployment manifests

Estimated MVP cost envelope

Expected infrastructure cost: approximately **$1,100–$1,600 per month**, with the GPU as the largest variable. In lower-utilization scenarios, actual spend may be closer to the lower end if scale-to-zero is working as intended.

Category	Estimated monthly cost	Comment
Always-on base infrastructure	$450–$850	AKS standard node, Service Bus, Blob, MongoDB, PostgreSQL, ACR, Key Vault, DNS
GPU usage	$200–$800	Depends on scan volume and spot pricing availability

Total MVP envelope	$1,100–$1,600	Suitable for pilot-scale usage

Recommendation

The MVP should use a **single Azure-based AKS deployment with one always-on standard node and one queue-triggered spot GPU node pool**, backed by **Service Bus, Blob Storage, MongoDB Atlas, Temporal with PostgreSQL, ACR, Key Vault, NGINX Ingress, KEDA, and Flux CD**. This is the minimum infrastructure set that can support the full Scan2BIM workflow reliably enough for early delivery while keeping costs controlled and preserving a straightforward path to production hardening later.